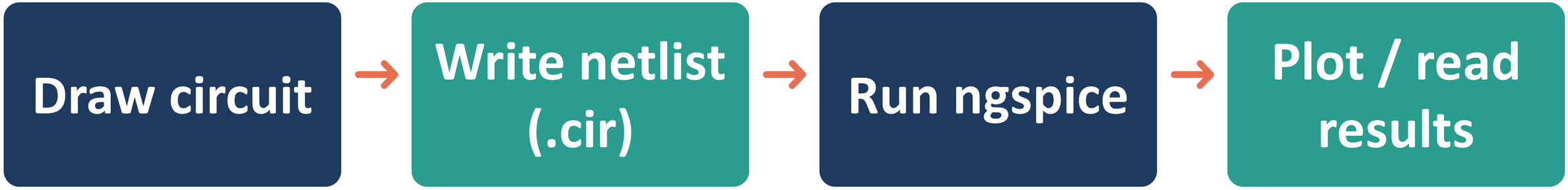
ngspice Basics

circuit to netlist to simulation

What is ngspice?

ngspice is a free, open-source circuit simulator. You describe a circuit as text (a netlist) and it calculates voltages and currents for you.



Why simulate? Test a design safely, cheaply and quickly before building it on a breadboard.

Components & naming conventions

Component	Letter	Example line
Resistor	R	`R1 1 2 1k`
Capacitor	C	`C1 2 0 1u`
Inductor	L	`L1 2 3 10m`
Diode	D	`D1 2 3 1N4148`
BJT transistor	Q	`Q1 3 2 0 2N2222`
JFET	J	`J1 1 2 0 JMOD`
MOSFET	M	`M1 d g s b NMOS1`
DC / AC source	V	`V1 1 0 DC 5`
Current source	I	`I1 1 0 DC 1m`
IC (subcircuit)	X	`X1 in out vcc 0 LM358`

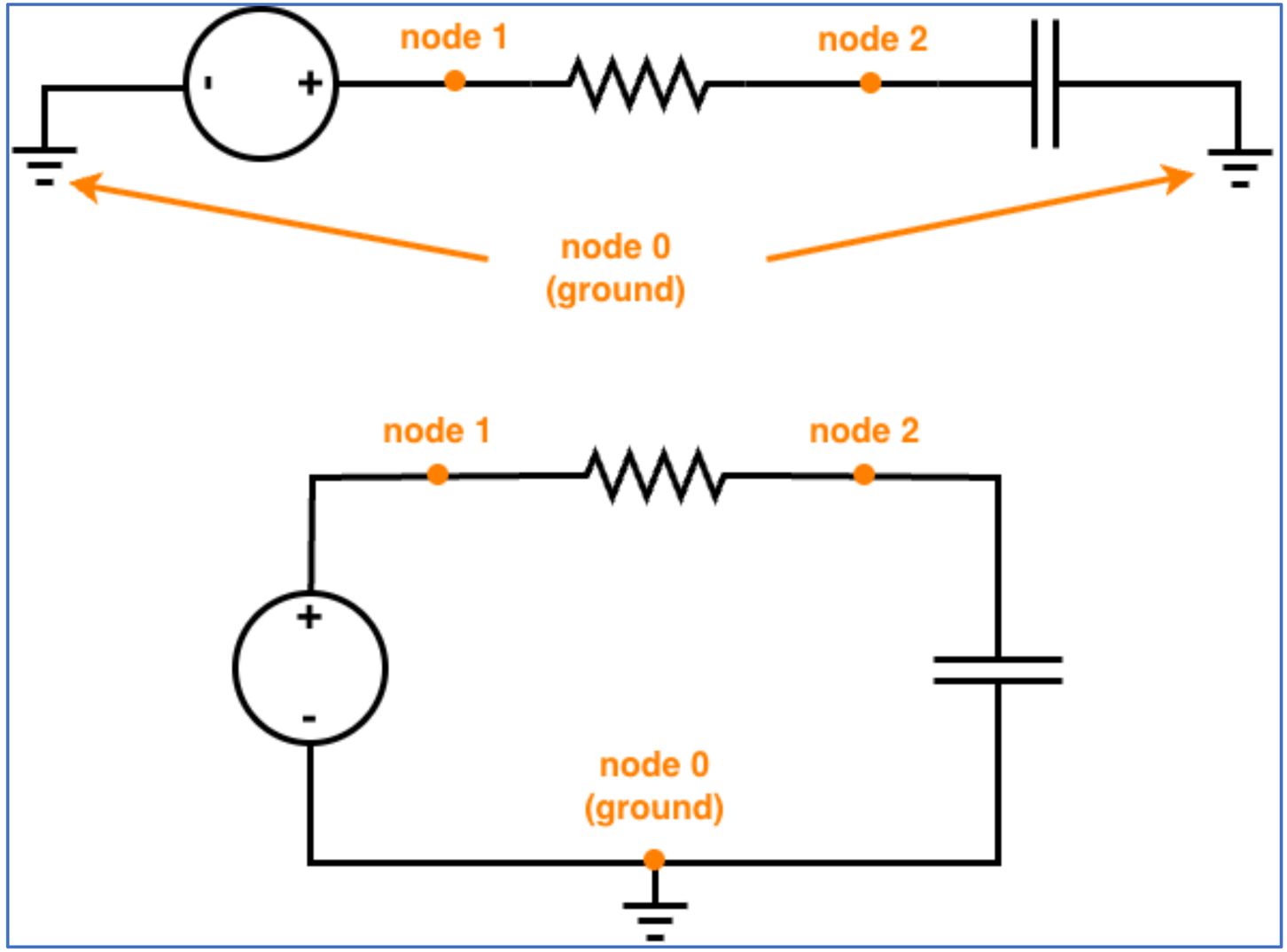
Rules

1st letter = type (R, C, D…)

Name = letter + label
: R1, Rload

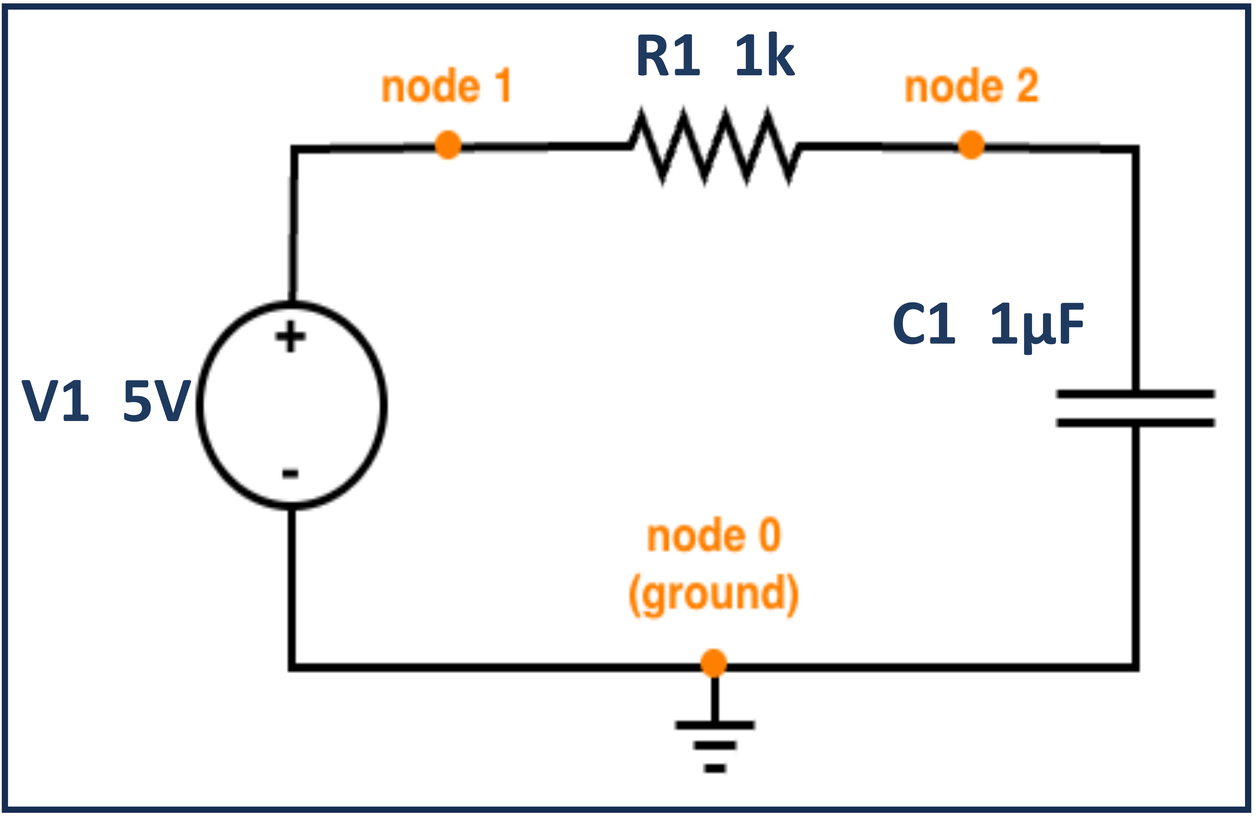
Values: k, m, u, n, p, MEG

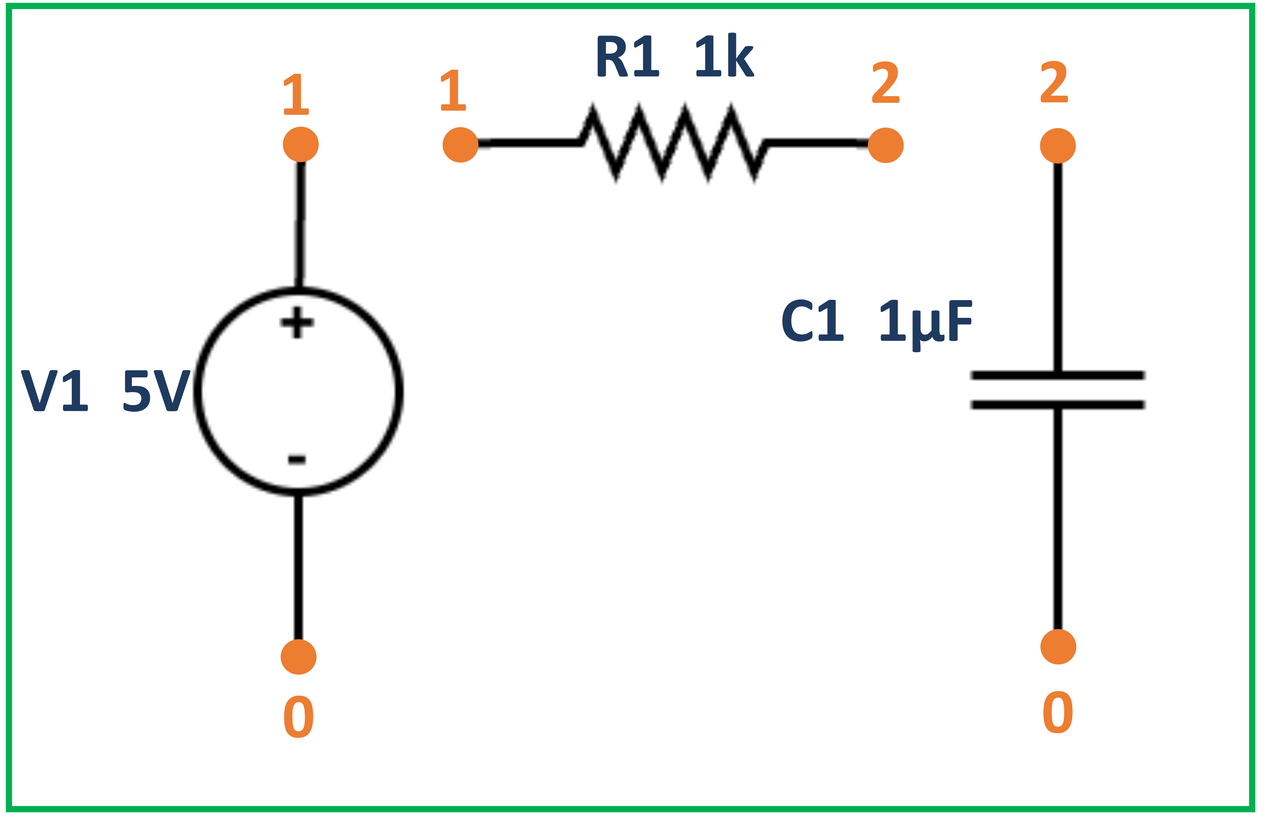
Nodes: the connection points



- A node is any point where components are connected by wire
- All points joined by plain wire are the SAME node
- Node 0 is always ground – the reference (0 V). Every circuit needs one
- Nodes can be numbers (1, 2, 3) or names (in, out, vcc)

Circuit → netlist (step by step)

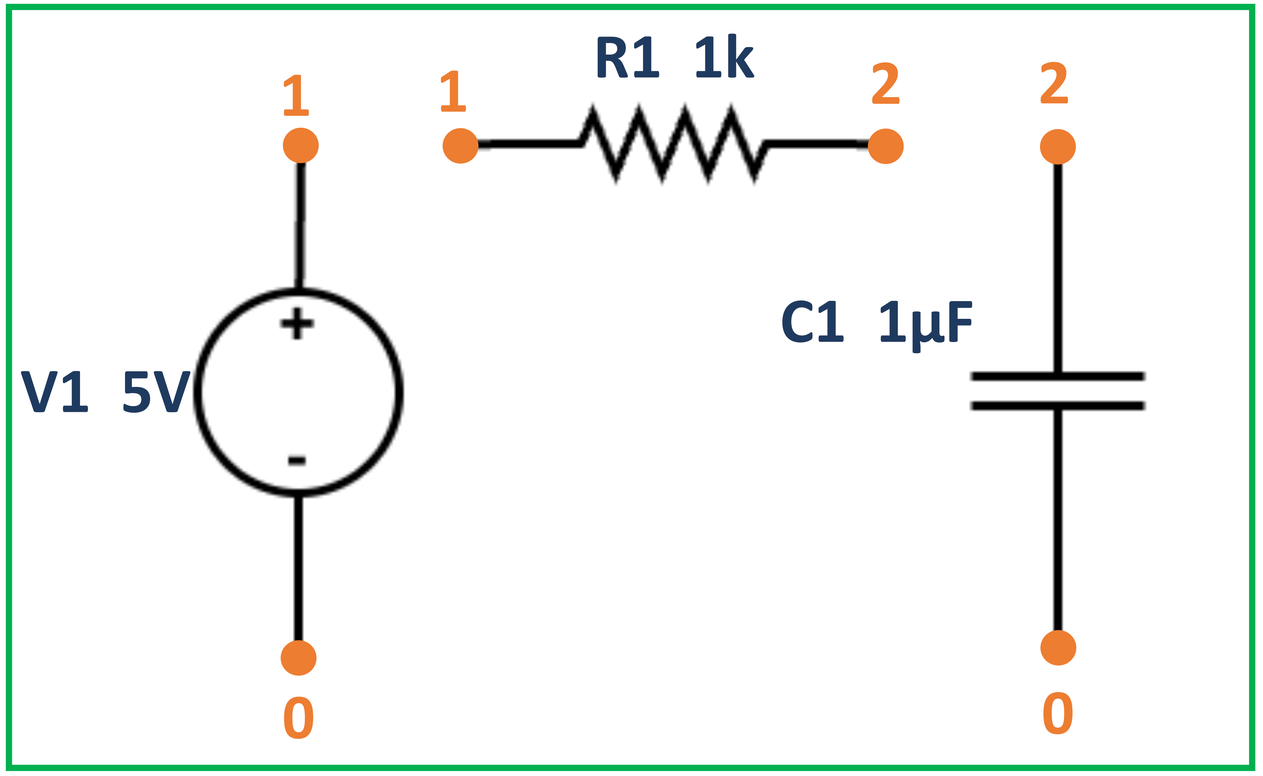




Circuit → netlist (step by step)

Steps

1. Input (V1) 5V DC across 1 and 0
2. Resistor (R1) across 1 and 2
3. Capacitor (C1) across 2 and 0
4. We need to plot input & output:



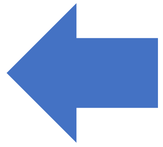
Input across 1 and 0 → v(1) ⇒ Voltage at 1 w.r.t ground (0)

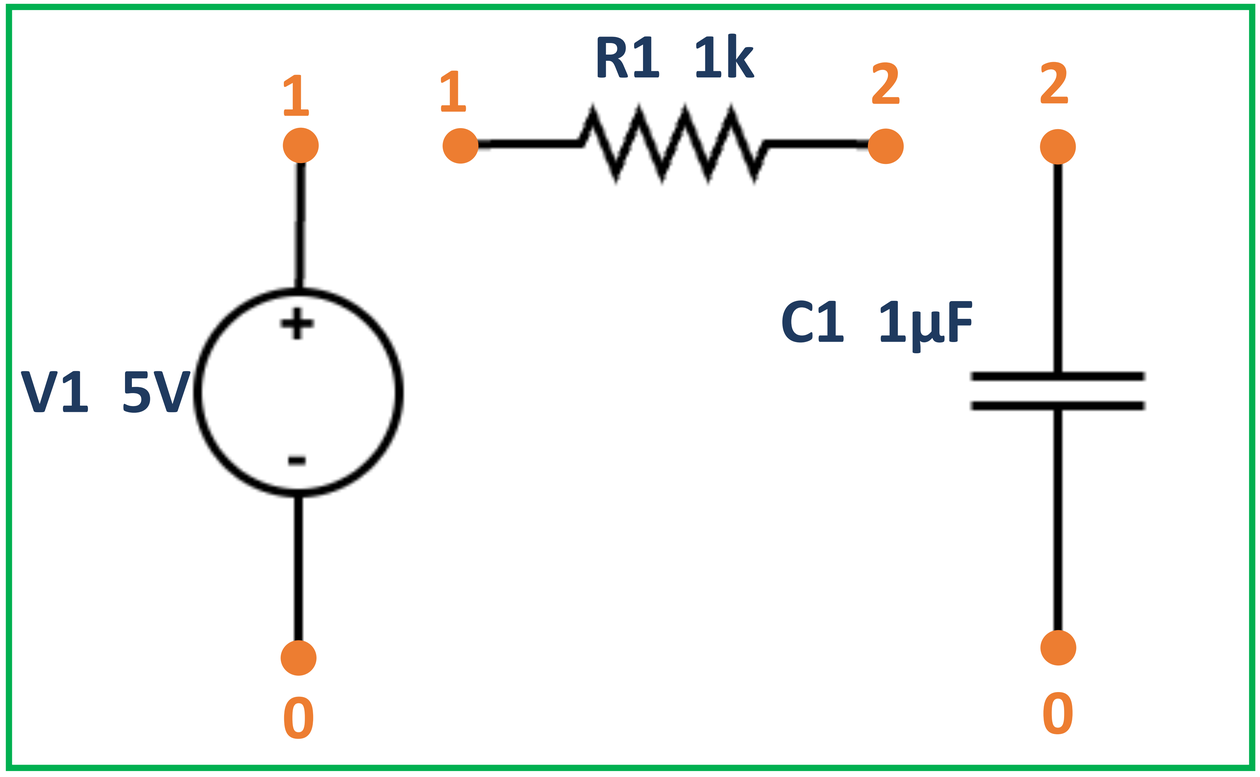
Output across 2 and 0 → v(2) ⇒ Voltage at 2 w.r.t ground (0)

Voltage across R1 ⇒ 1 and 2 → v(1, 2) ⇒ Voltage at 1 w.r.t reference (2)

Circuit → netlist (step by step)

```
RC low-pass
V1 1 0 DC 5
R1 1 2 1k
C1 2 0 1u
.op
.end
```





1. Number the nodes (ground = 0)
2. Name each component
3. One line per component:
   name node node value
4. Add an analysis + .end

Passive components: R, C, L

Resistor

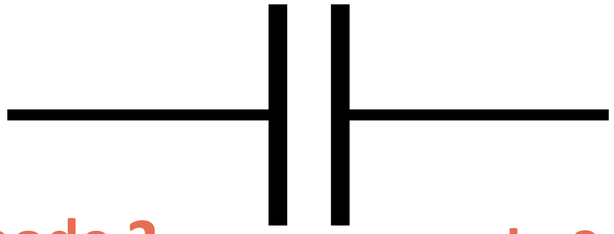


node 1 node 2

```
R1 1 2 1k
```

2 nodes, value in ohms

Capacitor

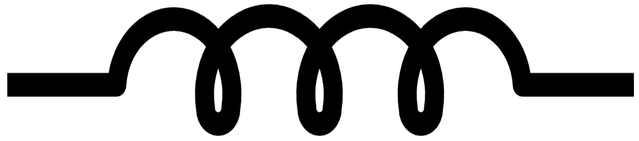


node 2 node 0

```
C1 2 0 10u
```

2 nodes, value in farads

Inductor



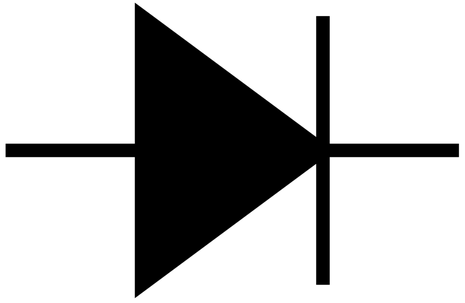
node 2 node 3

```
L1 2 3 10m
```

2 nodes, value in henries

Diode and transistor

Diode



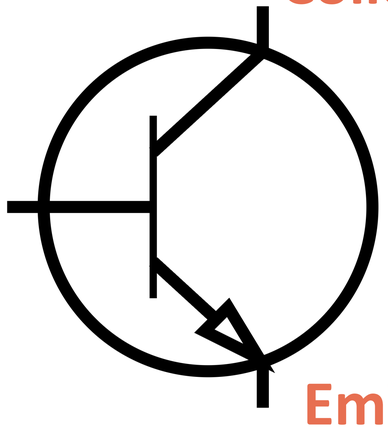
anode (1)

cathode (2)

```
D1 1 2 D1N4148
.model D1N4148 D(IS=1e-14)
```

Order: anode, cathode, model name

BJT (NPN)



Collector (3)

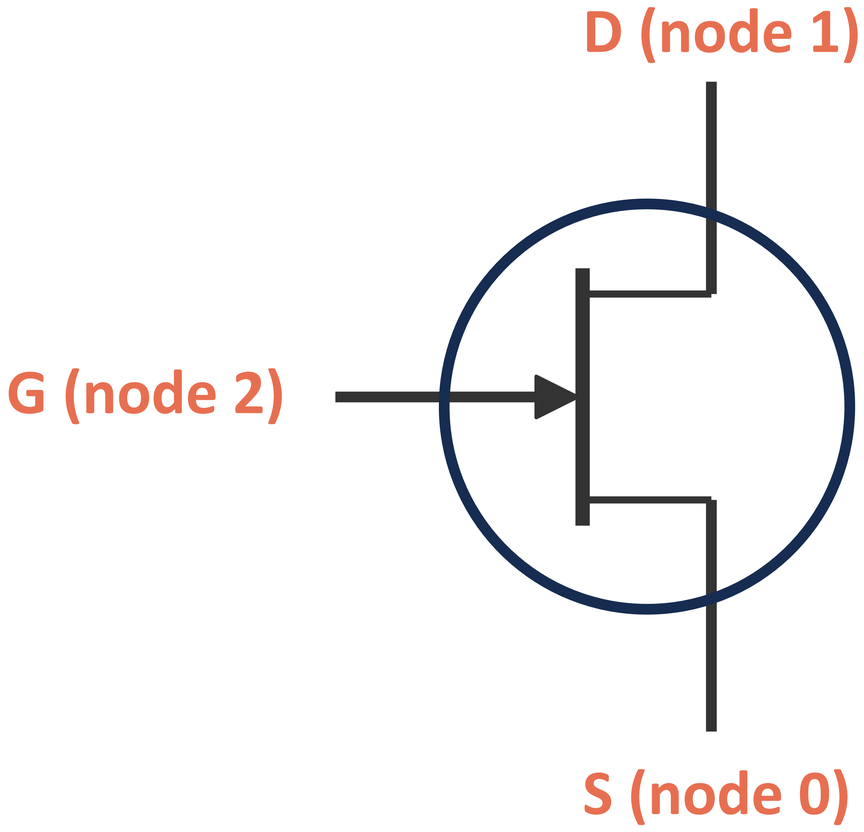
Base (2)

Emitter (0)

```
Q1 3 2 0 2N2222
.model 2N2222 NPN(BF=200)
```

Order: Collector, Base, Emitter, model

JFET (Junction Field-Effect Transistor)

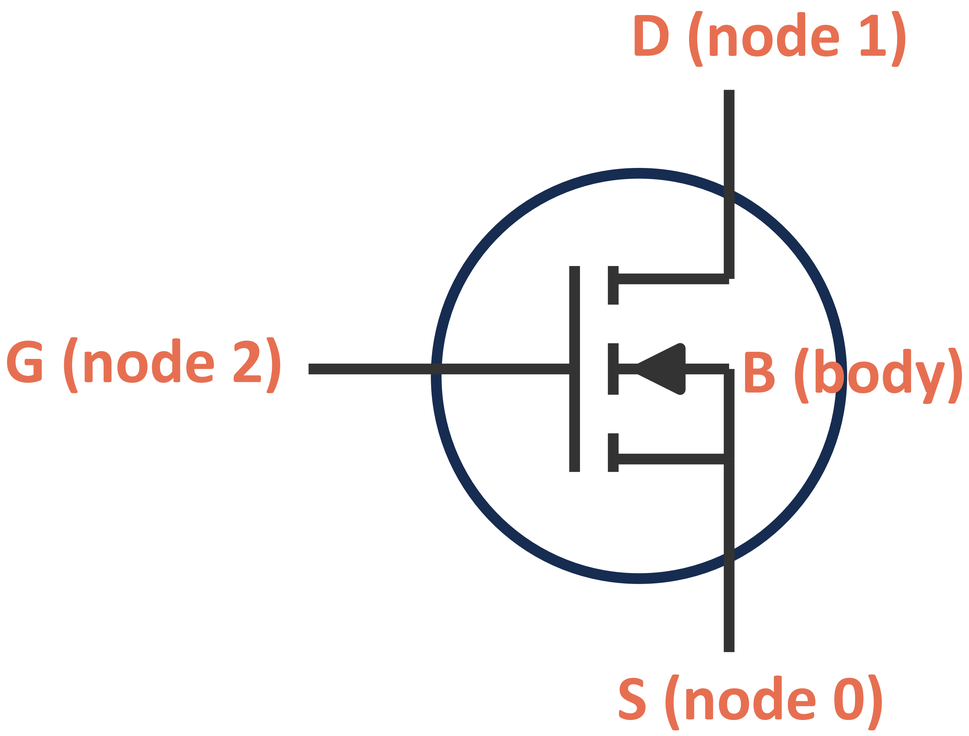


N-channel JFET symbol

```
J1 1 2 0 JMOD
.model JMOD NJF(VTO=-2 BETA=1m)
```

- Gate voltage controls the drain current
- N-channel JFET is ON at VGS = 0 and turns OFF at a negative VGS (pinch-off, VTO)
- Gate draws almost no current
- Pin order: Drain, Gate, Source
- P-channel: use PJF

MOSFET

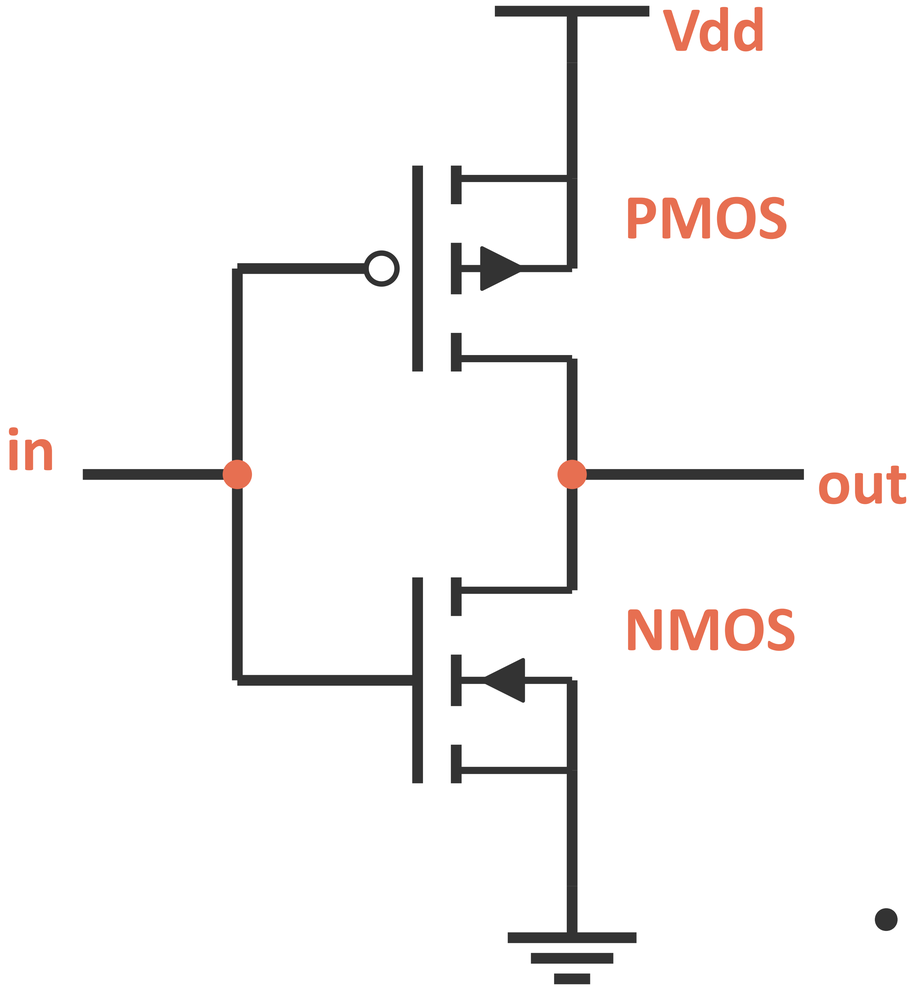


N-channel enhancement MOSFET

```
M1 1 2 0 0 NMOD W=10u L=1u
.model NMOD NMOS(VTO=1 KP=100u)
```

- Insulated gate: ON when VGS is above the threshold VTO
- Pin order: Drain, Gate, Source, Body
- Body is usually tied to the source
- W and L (width, length) set the size
- PMOS: use a PMOS model with VTO negative

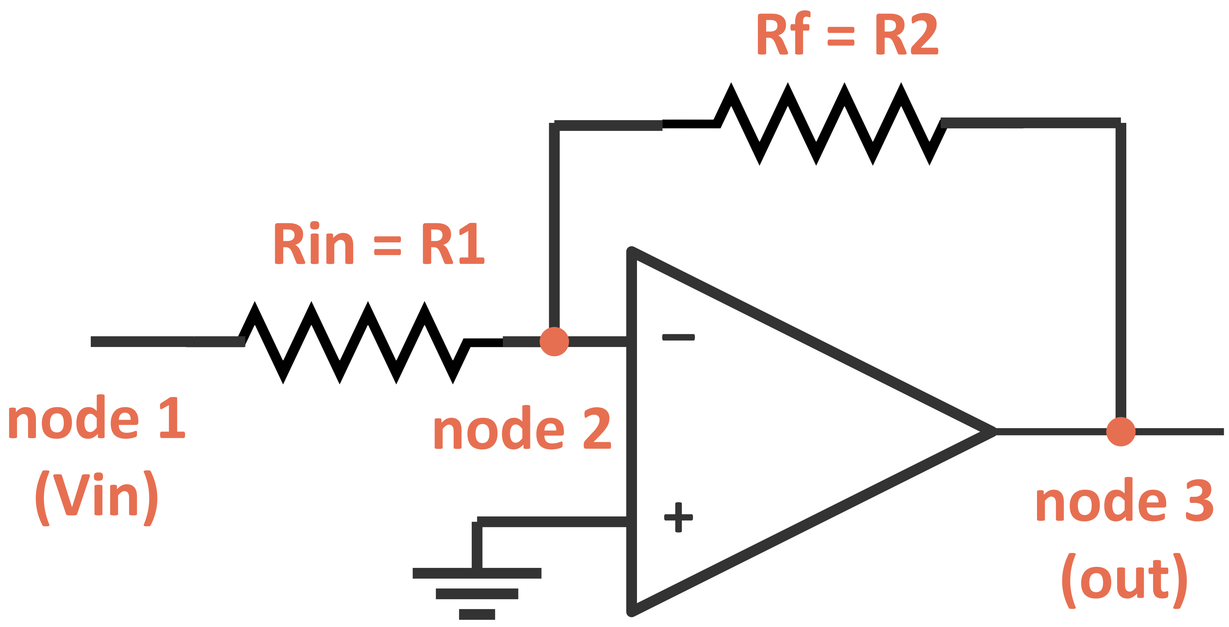
CMOS inverter (NOT gate)




```
VDD vdd 0 DC 5
VIN in 0 DC 0
MP1 out in vdd vdd PMOD
MN1 out in 0 0 NMOD
.model NMOD NMOS(VTO=1 KP=100u)
.model PMOD PMOS(VTO=-1 KP=50u)
.dc VIN 0 5 0.1
```

- in = 0 V: PMOS ON, NMOS OFF, so out = 5 V
- in = 5 V: PMOS OFF, NMOS ON, so out = 0 V
- Almost no power is used when the input is steady

Op-amp: inverting amplifier



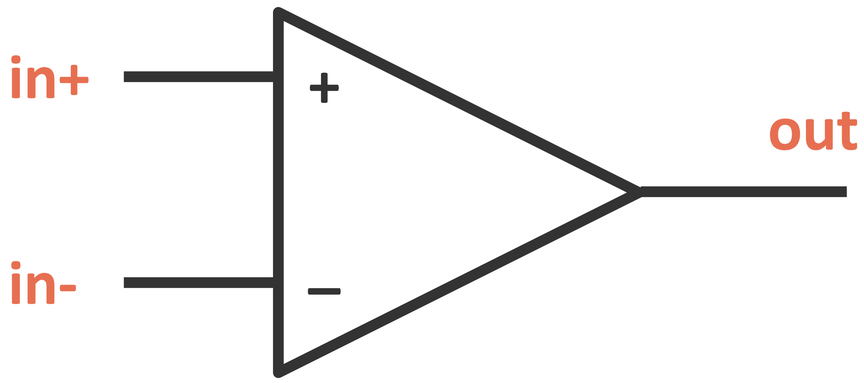
$$\text{Gain} = -\frac{R_f}{R_1} = -\frac{R_2}{R_1} = -10$$

```
Vin 1 0 DC 1
R1 1 2 1k
R2 2 3 10k
X1 0 2 3 OPAMP
.subckt OPAMP inp inn out
Rin inp inn 1MEG
E1 a 0 inp inn 1e5
Ro a out 50
.ends OPAMP
.op
.end
```

- E source = voltage-controlled voltage source (gain 1e5)
- Pin order: in+, in-, out. Output here: –10 V

ICs: subcircuits (X)

An IC (op-amp, gate, regulator…) is a small circuit packaged as a .subckt. You place it with the letter X and list its pins in order.

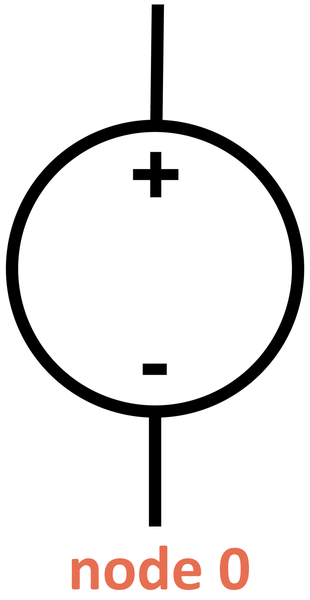


X1 (op-amp)

```
.include opamp.lib
X1 3 2 4 0 5 LM358
* pins:  in+ in- out ...  (check the
* model’s .subckt line for the order)
```

Power supplies: DC and AC

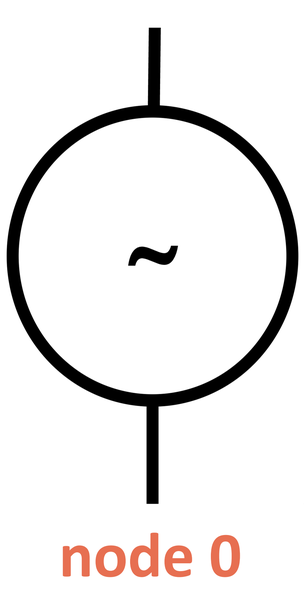
DC voltage source




```
V1 1 0 DC 5
```

Constant 5 V between node 1 and node 0

AC (sine) source




```
V2 1 0 SIN(0 5 1k)
```

Sine: offset 0 V, amplitude 5 V, 1 kHz

Other useful forms:

- I1 1 0 DC 1m (current source)
- V3 1 0 PULSE(0 5 0 1n 1n 1m 2m) (square wave)
- V4 1 0 AC 1 (for AC sweep)

Analysis commands

.op

DC operating point: all node voltages at one instant

```
.op
```

.dc

Sweep a source, e.g. 0 to 5 V

```
.dc V1 0 5 0.1
```

.ac

Frequency response (Bode plot)

```
.ac dec 10 1 100k
```

.tran

Voltages vs time

```
.tran 0.1m 10m
```

Running ngspice

```
$ ngspice rc.cir

ngspice 1 -> run
ngspice 2 -> print v(2)
ngspice 3 -> plot v(1) v(2)
ngspice 4 -> quit
```

Steps

1. Save your netlist as rc.cir
2. Open a terminal, run ngspice rc.cir
3. Type run to simulate
4. print v(node) shows values;

   plot v(node) draws a graph
5. v(2) = voltage at node 2;

   i(V1) = current through V1

Question

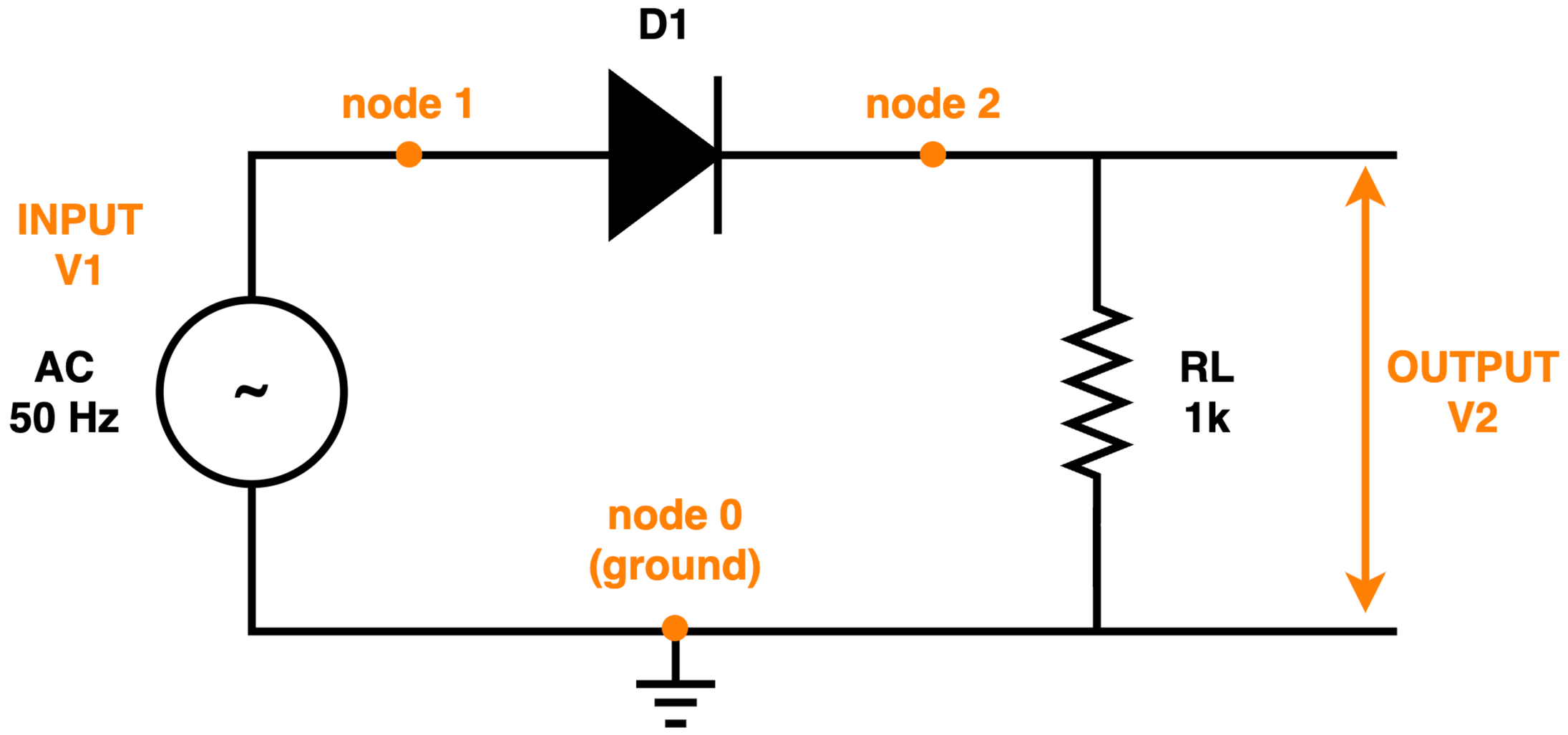
1. Design half-wave and full-wave rectifiers for an AC source with amplitude 10 and frequency 50 Hz, plot the input sine wave and the output pulse
2. Is the output DC? Add a capacitor (1000 µF, at output) and see the changes

Report Format

1. Title
2. Aim / Objective
3. Components Required
4. Circuit Diagram (with all nodes labeled)
5. Working Principle
6. ngspice Netlist
7. Simulated Waveforms / Bode Plots
8. Observation and Results

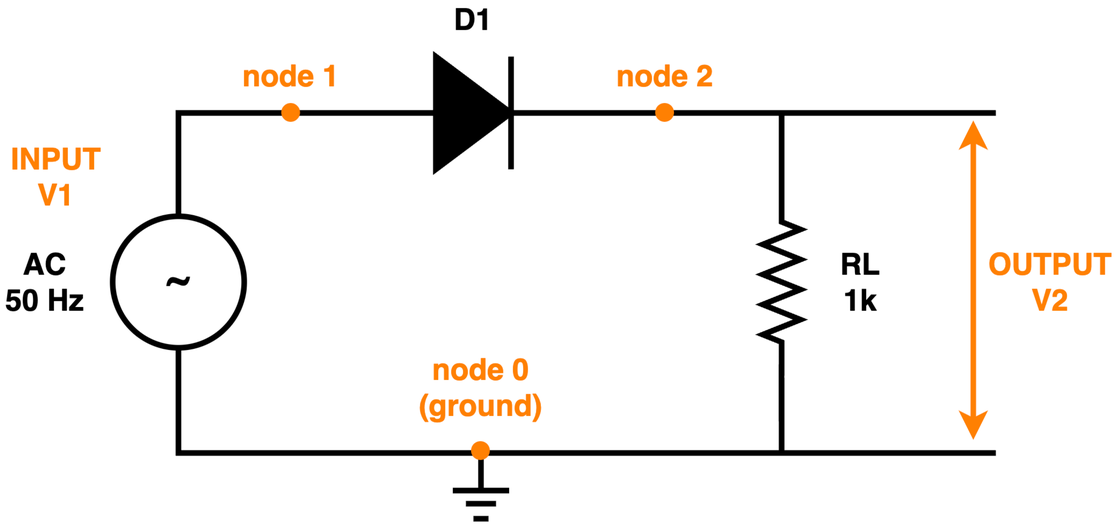
Half-wave Rectifier

Circuit



Half-wave Rectifier

Circuit



1. Input (V1) sin wave across 1 and 0
2. Diode (D1) across 1 and 2
3. Load resistor (RL) across 2 and 0
4. We need to plot input & output:

   Input across 1 and 0

   Output across 2 and 0
5. plot v(1) ⇒ Voltage at 1 with respect to ground (0) no need to give v(1,0)
6. plot v(2) ⇒ Voltage at 2 with respect to ground (0) no need to give v(2,0)

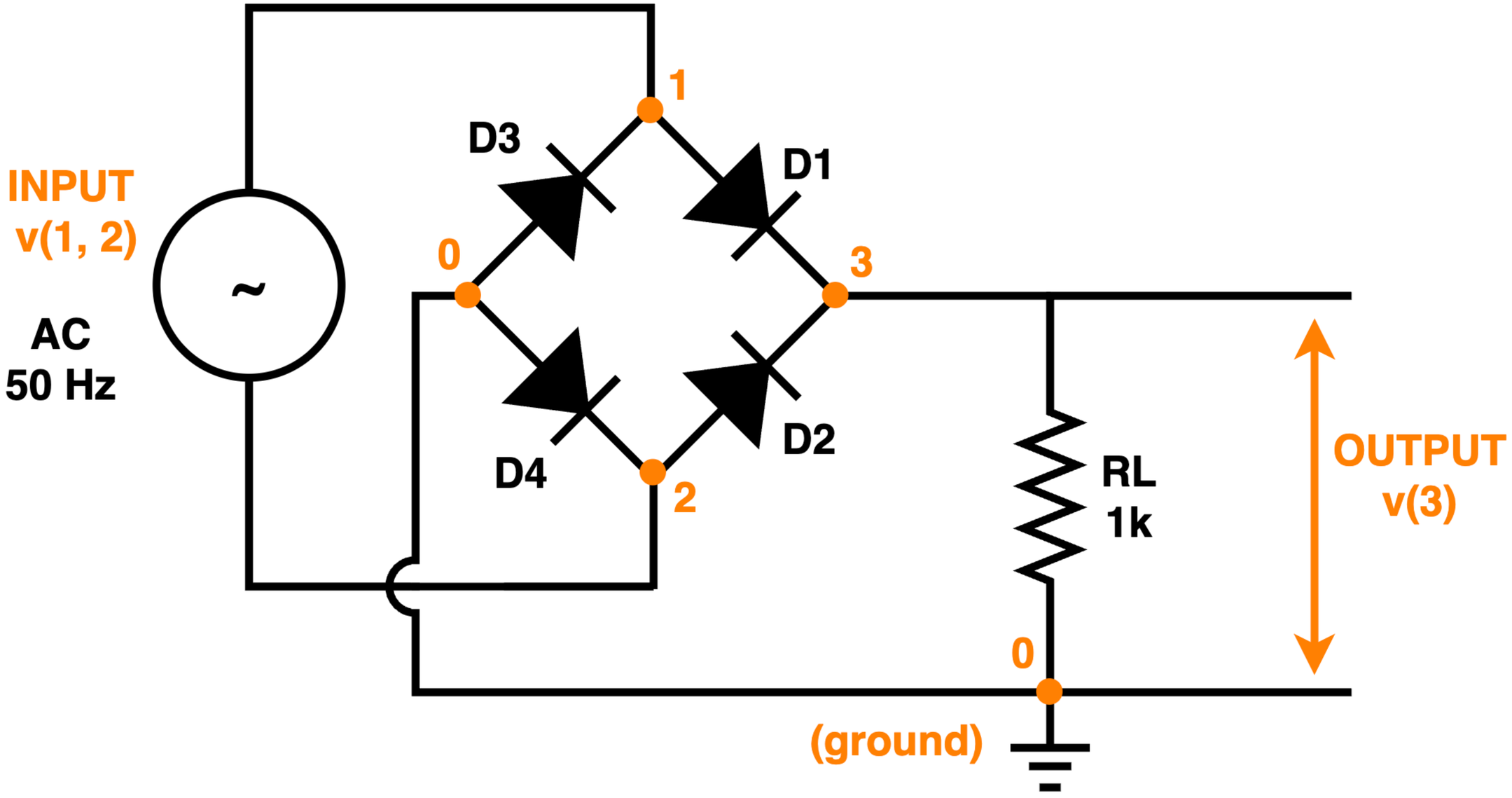
Half-wave Rectifier

Netlist

```
* Half-wave rectifier
V1 1 0 SIN(0 10 50)
D1 1 2 Dmod
RL 2 0 1k
.model Dmod D(Is=2.52n N=1.7)
.control
set noaskquit
tran 100u 100m
plot v(1) v(2) title "Half-Wave Rectifier"
.endc
.end
```

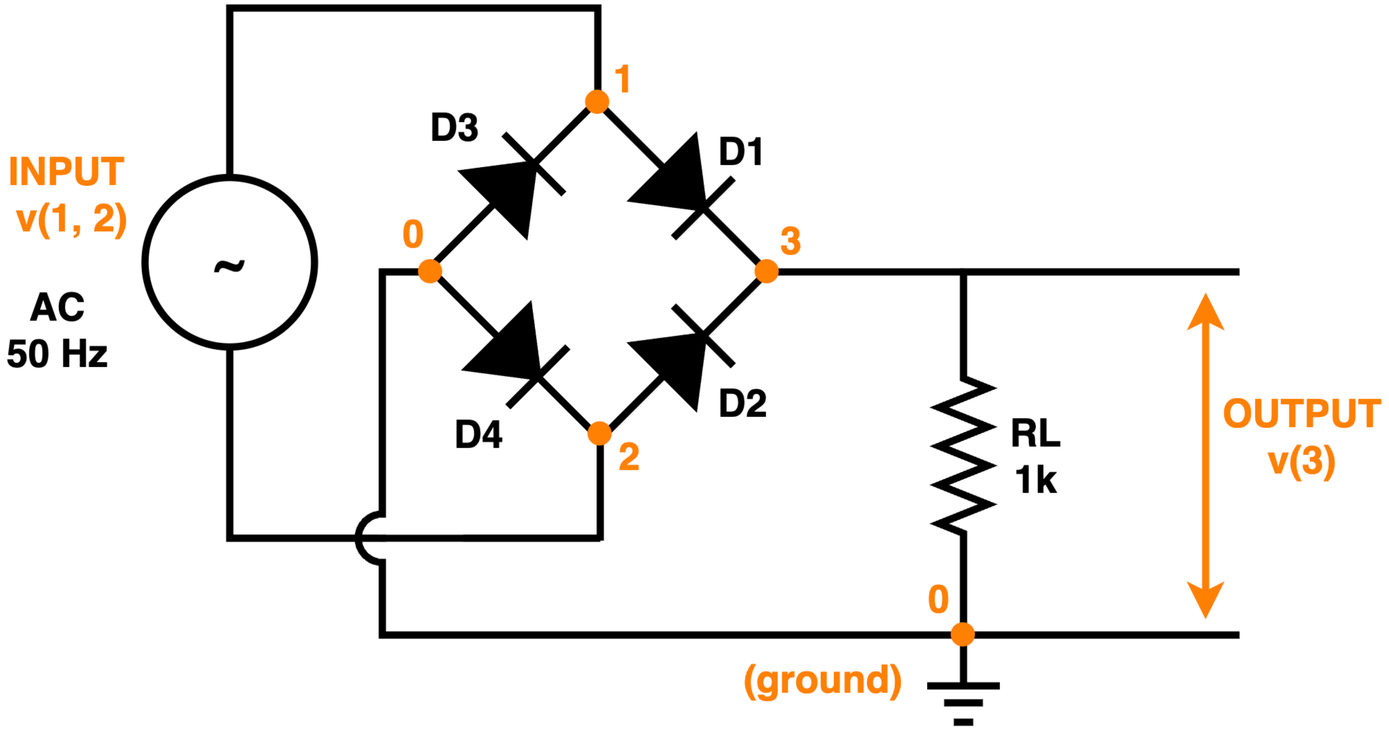
Full-wave Rectifier

Circuit



Half-wave Rectifier

Circuit



1. Input (V1) sin wave across 1 and 2
2. D1 across 1 and 3
3. D2 across 2 and 3
4. D3 across 0 and 1
5. D4 across 0 and 2
3. Load resistor (RL) across 3 and 0
4. We need to plot input & output:

   Input across 1 and 2 ⇒ v(1, 2)

   Output across 3 and 0 ⇒ v(3)

Half-wave Rectifier

Netlist

```
* Full-wave bridge rectifier
V1 1 2 SIN(0 10 50)
D1 1 3 Dmod
D2 2 3 Dmod
D3 0 1 Dmod
D4 0 2 Dmod
RL 3 0 1k
.model Dmod D(Is=2.52n N=1.7)
.control
set noaskquit
tran 100u 100m
plot v(1,2) v(3) title "Full-Wave Bridge Rectifier"
.endc
.end
```

Common mistakes & summary

Avoid these

- No node 0 (no ground)
- Floating node (only one connection)
- Missing .end
- Writing M for mega (use MEG)
- Wrong pin order for D, Q, X
- First line is the title – not a component!

Remember

- Node 0 = ground
- First letter = component type
- One line per component
- Netlist = name, nodes, value
- Add an analysis (.op/.dc/.ac/.tran)
- End with .end